

Algebraic Topology

Lectured by Steven Sivek

Scribed by Yu Coughlin

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1 Covering spaces

Definition 1.0.1. A space X is **locally connected** if for $x \in X$ and $x \in V \subseteq X$ open, there exists an open $x \in U \subset V$ which is connected, we can do the same for path-connected.

For this entire section, all spaces are assumed to be path-connected and locally path-connected.

Definition 1.0.2. A continuous map $p : \tilde{X} \rightarrow X$ is **covering map** if for all $x \in X$, there exists an open neighborhood $x \in U \subset X$ such that

$$p^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$$

is a nonempty disjoint union of open $V_{\alpha} \subset \tilde{X}$, with $p|_{V_{\alpha}} : V_{\alpha} \rightarrow U$ a homeomorphism for all α .

Lemma 1.0.3. Covering maps are surjective.

1.1 Lifting

Lemma 1.1.1. If $p : \tilde{X} \rightarrow X$ is a covering space, with $\tilde{f}_0, \tilde{f}_1 : Y \rightarrow \tilde{X}$ two lifts of $f : Y \rightarrow X$, then the set

$$S = \{y \in Y \mid \tilde{f}_0(y) = \tilde{f}_1(y)\}$$

is both open and closed in Y .

Proof. For all $y \in S$, there exists an open neighbourhood $f(y) \in U \subset X$ with $p^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$, then $\tilde{f}_0(y) = \tilde{f}_1(y) \in V_{\beta}$, and thus on $N := \tilde{f}_0^{-1}(V_{\beta}) \cap \tilde{f}_1^{-1}(V_{\beta})$ we have

$$p|_{V_{\beta}} \circ \tilde{f}_0|_N = f|_N = p|_{V_{\beta}} \circ \tilde{f}_1|_N$$

but $p|_{V_{\beta}}$ is a homeomorphism, so $\tilde{f}_0|_N = \tilde{f}_1|_N$, with $y \in N \subset S$ open, thus S is open.

For all $y \in \bar{S} \setminus S$, there exists an open neighbourhood $f(y) \in U \subset X$ with $p^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$, and thus $\tilde{f}_0(y) \in V_{\beta}$ and $\tilde{f}_1(y) \in V_{\gamma}$ for $\beta \neq \gamma$, so the open neighbourhood $\tilde{f}_0^{-1}(V_{\beta}) \cap \tilde{f}_1^{-1}(V_{\gamma})$ of y intersects S , but then $V_{\beta} \cap V_{\gamma} \neq \emptyset$, a contradiction. $\square$

Theorem 1.1.2 (Homotopy lifting theorem). For $p : \tilde{X} \rightarrow X$ a covering map, and Y a locally connected space, for all homotopies $H : Y \times I \rightarrow X$ with a lift $\tilde{h}_0 : Y \times \{0\} \rightarrow \tilde{X}$ of $h_0(y) = H(y, 0)$, there exists a unique homotopy

$$\begin{array}{ccc} Y \times \{0\} & \xrightarrow{h_0} & \tilde{X} \\ \downarrow & \exists! \tilde{H} \nearrow & \downarrow p \\ Y \times I & \xrightarrow{H} & X \end{array}$$

Proof. For each $y \in Y$, consider the map $\gamma_y : I \rightarrow X$ by $\gamma_y(t) = H(y, t)$, if we have two lifts $\tilde{H}, \tilde{K}$ of H , then they agree at $(y, 0)$, and thus by the previous lemma, as I is connected, they agree on all of I , hence $\tilde{H} = \tilde{K}$.

Now, fix $y_0 \in Y$, for all $t \in I$, there exists open $y_0 \in V_t \subset Y$ and $t \in W_t \subset I$, such that $H(V_t \times \overline{W_t})$ lies in a single evenly covered U_{α} , as I is compact, we can cover this with finitely many W_t and let V be the intersection of the corresponding V_t . As Y is locally connected, we may shrink V to be connected and consider the $H(V \times [t_i, t_{i+1}])$ each contained in some U_{α_i} for $i = 0, 1, \dots, n-1$; now, starting with $\tilde{h}_0$, we can extend each $\tilde{H}(V \times [0, t_i])$ to $\tilde{H}(V \times [0, t_{i+1}])$ by taking the unique lift of the connected $V \times [0, t_{i+1}]$ lying entirely in the same V_{β_i} over U_{α_i} as $V \times \{t_i\}$.

We now have, for all $y \in Y$, a lift $\tilde{H}_y : V_y \times I \rightarrow \tilde{X}$, but by the previous lemma, these all agree on all their intersections, so we can glue these all together to $\tilde{H} : Y \times I \rightarrow \tilde{X}$. $\square$

Corollary 1.1.3 (Path lifting theorem). For $p : \tilde{X} \rightarrow X$ a covering map, and $\gamma : I \rightarrow X$ a path, then for each $\tilde{x} \in p^{-1}(\gamma(0))$, there exists a unique path $\tilde{\gamma} : I \rightarrow \tilde{X}$ lifting γ to $\tilde{X}$ and starting at $\tilde{x}$.

Proof. This is the homotopy lifting theorem with $Y = *$. $\square$

1.2 Covers and the fundamental group

Definition 1.2.1. For (X, x) a pointed topological space, let $\mathbf{Cov}(X, x)$ be the full subcategory of $\mathbf{Top}_*/(X, x)$ of covering maps. We get the functor $\pi_1 : \mathbf{Cov}(X, x) \rightarrow \mathbf{Grp}$.

Proposition 1.2.2. If $p : (Y, y) \rightarrow (X, x)$ is a covering map, then $p_* : \pi_1(Y, y) \rightarrow \pi_1(X, x)$ is injective.

Proof. If $\gamma : I \rightarrow Y$ has $p_*([\gamma]) = [e_x]$, in X , then the homotopy $p \circ \gamma \simeq e_x$ has a unique lift to $\gamma \simeq e_y$. $\square$

So can associate to each cover $p : Y \rightarrow X$ a subgroup $p_*\pi_1(Y, y) \leq \pi_1(X, x)$.

Theorem 1.2.3 (Lifting criterion). If $p : (Y, y) \rightarrow (X, x)$ is a covering map and $f : (Z, z) \rightarrow (X, x)$ continuous with Z path-connected and locally path-connected, then we can lift f iff $f_*\pi_1(Z, z) \subseteq p_*\pi_1(Y, y)$.

Proof. The $(\Rightarrow)$ direction is obvious, so we suppose $f_*\pi_1(Z, z) \subseteq p_*\pi_1(Y, y)$, and define the lift $\tilde{f}$. For all $w \in Z$, there exists a path $\gamma : I \rightarrow Z$ from z to w , we can lift $f \circ \gamma$ to some unique g with $g(0) = y$, now take $\tilde{f}(w) := g(1)$.

First, to show $\tilde{f}$ is well-defined, suppose we have another path $\gamma' : I \rightarrow Z$ from z to w , then consider $\phi = f \circ (\gamma * \bar{\gamma}') \in f_*\pi_1(Z, z) \subseteq p_*\pi_1(Y, y)$, so we can find a loop $\psi : I \rightarrow Y$ based at y with $p_*[\psi] = [\phi]$, so there is a homotopy H from $p \circ \psi$ to ϕ in X , which uniquely lifts to some $\tilde{H}$ from ψ to some lift $\tilde{\phi}$ of ϕ , which must be a loop based at y . But, we can form another lift of $\tilde{\phi}$ by following g , and then g' , a lift of $\tilde{f} \circ \gamma'$ starting at $g(1)$, by uniqueness, this lift must be a loop at y , so the unique lift of $f \circ \gamma'$ must start at y and end of $g(1)$, so $\tilde{f}(w)$ doesn't depend on our choice of path.

Now, to show $\tilde{f}$ is continuous, let $U \subset X$ be an open, path-connected, evenly covered neighbourhood of $f(w)$, with V_α the unique sheet containing $\tilde{f}(w)$, as the U, V form a basis of X, Y respectively, it suffices to show $\tilde{f}^{-1}(V)$ contains an open neighbourhood of w . Let $W \subset f^{-1}(U) \subset Z$ be a path-connected open neighbourhood of w , then for all $w' \in W$, there exists a path $\eta : I \rightarrow W$ from w to w' , its image $f \circ \eta$ lies entirely within U , where p is a homeomorphism from V , so we have the lift $p^{-1} \circ f \circ \eta$ whose starting point is $\tilde{f}(w)$, by the uniqueness of lifts, the endpoint of $p^{-1} \circ f \circ \eta \in V$ is $\tilde{f}(w')$, therefore $W \subset \tilde{f}^{-1}(V)$, hence $\tilde{f}$ is continuous. $\square$

Corollary 1.2.4. If $p : (Y, y) \rightarrow (X, x)$ and $q : (Z, z) \rightarrow (X, x)$ are two non-isomorphic coverings of (X, x) , then $p_*\pi_1(Y, y) \neq q_*\pi_1(Z, z)$.

Proof. Suppose $p_*\pi_1(Y, y) = q_*\pi_1(Z, z)$, then the lifting criterion gives $f : (Y, y) \rightarrow (Z, z)$ and $g : (Z, z) \rightarrow (Y, y)$, but now $f \circ g$ and id_Y are both lifts of p along p , and so by uniqueness are equal. $\square$

Proposition 1.2.5. If in the statement of the lifting criterion, $f = q$ is a covering map, then the lift r is a covering map.

Proof. Take $v \in Y$, then $p(v) \in X$ has evenly open neighbourhoods U_y and U_z by p and q respectively, take $U \subset U_y \cap U_z$ connected, and have its sheets be V_α and W_β in p and q respectively, let V_0 be the sheet containing v . For all $r(w) \in V_0$, with $w \in Z$, we have $q(w) = pr(w) \in U$, so $w \in W_\beta$ for some β , as W_β is a connected, and $r(W_\beta)$ is a subset of $p^{-1}(U)$, we must have $r(W_\beta) \subset V_0$, with $r|_{W_\beta} = (p|_{V_0})^{-1} \circ q|_{W_\beta}$ a homeomorphism, so $r^{-1}(V_0)$ is a disjoint union of some of the W_β , with each $r|_{W_\beta}$ a homeomorphism onto V_0 , hence r is a covering map. $\square$

1.3 Deck transformations

Theorem 1.3.1. If $p : Y \rightarrow X$ is a covering map, then for each $x \in X$, there is a right action $p^{-1}(x) \curvearrowright \pi_1(X, x)$. For all $y \in p^{-1}(x)$ and $\gamma : I \rightarrow X$, let $\tilde{\gamma}$ be the unique lift of γ to Y with starting point y , then take $y \cdot [\gamma] = \tilde{\gamma}(1) \in p^{-1}(x)$.

Proof. To show this is well-defined, suppose γ, γ' are loops at x in X , with H a homotopy from γ to γ' , then there is a unique lift

$$\begin{array}{ccc} I \times \{0\} & \xrightarrow{\tilde{\gamma}} & Y \\ \downarrow & \tilde{H} \nearrow & \downarrow p \\ I \times I & \xrightarrow{H} & X \end{array}$$

which must satisfy $\tilde{H}(0, t) = \tilde{\gamma}(0) = y$ and $\tilde{H}(1, t) = \tilde{\gamma}(1)$ for all t , which implies $\tilde{\gamma}'(1) = \tilde{\gamma}(1)$, so $y \cdot [\gamma] = y \cdot [\gamma']$.

To check this is a group homomorphism, we have to show $(y \cdot [\gamma]) \cdot [\gamma'] = y \cdot ([\gamma] \cdot [\gamma'])$, let $\tilde{\gamma}''$ be the unique lift of $\gamma * \gamma'$ starting at y , then the RHS is $\tilde{\gamma}''(1)$. But the first half of $\tilde{\gamma}''$ must be the unique lift of γ starting at y , and the second half the unique lift of γ' starting at $\tilde{\gamma}(1)$, so $\tilde{\gamma}''(1) = \tilde{\gamma}'(1)$. $\square$

As Y is path-connected, this action is clearly transitive.

Proposition 1.3.2. For $p : Y \rightarrow X$ a covering with $x \in X$, the stabiliser of $y \in p^{-1}(x)$, is $p_*\pi(Y, y)$.

Proof. We can lift any loop γ in the stabiliser to a loop $\tilde{\gamma}$ in Y , then $\gamma = p_*([\tilde{\gamma}])$. $\square$

By orbit-stabiliser, we define the **degree** $\deg(p) := |p^{-1}(x)| = [\pi_1(X, x) : p_*\pi_1(Y, y)]$.

Proposition 1.3.3. For $p : Y \rightarrow X$ a covering with $x \in X$ and $y \in p^{-1}(x)$, then for all $[\gamma] \in \pi_1(X, x)$, we have $p_*\pi_1(Y, y \cdot [\gamma]) = [\gamma]^{-1}p_*\pi_1(Y, y)[\gamma]$.

Proof. Let $\tilde{\gamma}$ be the lift of γ from y to z in Y , this induces an isomorphism $\phi_{\tilde{\gamma}} : \pi_1(Y, y) \rightarrow \pi_1(Y, z)$ by $[\psi] \mapsto [\tilde{\gamma} * \psi * \tilde{\gamma}]$, and so $p_*\phi_{\tilde{\gamma}}([\psi]) = [(p \circ \tilde{\gamma}) * (p \circ \psi) * (p \circ \tilde{\gamma})] = [\tilde{\gamma} * (p \circ \psi) * \tilde{\gamma}] = [\gamma]^{-1}p_*([\psi])[\gamma]$. $\square$

Definition 1.3.4. We call an isomorphism of $p : (Y, y) \rightarrow (X, x)$ in $\mathbf{Cov}(X, x)$ a **Deck transformation** and write $\text{Deck}(p)$ or $\text{Deck}(Y)$ when clear from context. If $\text{Deck}(Y)$ acts transitively on $p^{-1}(x)$, then we call p **normal**.

Proposition 1.3.5. For $p : Y \rightarrow X$ a covering, with $x \in X$, and $y, z \in p^{-1}(x)$, then there exists $f \in \text{Deck}(Y)$ with $f(y) = z$ iff $p_*\pi_1(Y, y) = p_*\pi_1(Y, z)$.

Proposition 1.3.6. Such a deck transformation is exactly a lift of p against p , so this is just the lifting criterion.

Theorem 1.3.7. A covering $p : (Y, y) \rightarrow (X, x)$ is regular iff $p_*\pi_1(Y, y) \trianglelefteq \pi_1(X, x)$.

Proof. This is combining all the previous props. $\square$

Theorem 1.3.8. For $p : (Y, y) \rightarrow (X, x)$ a covering, $\text{Deck}(Y) \cong N_{\pi_1(X, x)}(p_*\pi_1(Y, y))/p_*\pi_1(Y, y)$

Proof. Write $G = \pi_1(X, x)$ and $H = p_*\pi_1(Y, y)$, and define a map $\phi : N_G(H) \rightarrow \text{Deck}(Y)$ by for all $[\gamma] \in N_G(H)$, as $p_*\pi_1(Y, y \cdot [\gamma]) = [\gamma]^{-1}p_*\pi_1(Y, y)[\gamma] = p_*\pi_1(Y, y)$ which gives a unique deck transformation $f = \phi([\gamma])$ such that $f(y) = y \cdot [\gamma]$.

To show ϕ is a homomorphism, take $[\gamma], [\gamma'] \in N_G(H)$ with $\phi([\gamma]) = f$ and $\phi([\gamma']) = g$, then we can lift γ' to a path α from $y \cdot [\gamma]$ to $(y \cdot [\gamma]) \cdot [\gamma']$, then $f^{-1} \circ g$ is the unique lift of γ' starting at y , so $f^{-1}(g(y)) = y \cdot [\gamma']$, thus $\phi([\gamma] \circ [\gamma']) = f(g(y)) = y \cdot [\gamma'] = \phi([\gamma'])$.

For surjectivity, $f \in \text{Deck}(Y)$ sending y to $z \in p^{-1}(x)$, then the transitive $\pi_1(X, x)$ action on $p^{-1}(x)$ gives γ with $z = y \cdot [\gamma]$, so $p_*\pi_1(X, y) = p_*\pi_1(X, y \cdot [\gamma])$, hence γ is in the normaliser.

$[\gamma] \in N_G(H)$ has $\phi[\gamma] = \text{id}_Y$ iff $y \cdot [\gamma] = y$ iff $[\gamma] \in P_*\pi_1(Y, y) = H$. $\square$

1.4 Universal covers

The lifting criterion for covering spaces says is $(\tilde{X}, \tilde{x})$ is simply connected, then it is initial in $\mathbf{Cov}(X, x)$, the previous theorem thus says $\pi_1(X, x) \cong \text{Deck}(\tilde{X})$, we want to know when these exists.

Definition 1.4.1. A space X is **semi-locally simply connected** if all $x \in X$ has an open neighbourhood $U \subset X$ such that the induced map from the inclusion $i_*\pi_1(U, x) \rightarrow \pi_1(X, x)$ is trivial.

This is weaker than locally simply connected as we allow loops in our neighbourhood to be trivial only when considered in all of $\pi_1(X, x)$. Consider the cone on the Hawaiian earring for something semi-locally simply connected but not locally simply-connected

Theorem 1.4.2. If X is path-connected, locally path-connected, and semi-locally simply connected, then X admits a simply connected cover $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$.

Proof. First consider the set $\tilde{X} = \{\gamma : I \rightarrow X \mid \gamma(0) = x\}$ module homotopy rel the endpoints, and let $\tilde{x} = [e_x]$ and $p : \tilde{X} \rightarrow X$ send $[\gamma]$ to $\gamma(1)$, which is clearly well-defined.

To define a topology on this set, first consider the set $\mathcal{U}$ of path-connected open sets $U \subset X$ where $i_* : \pi_1(U, z) \rightarrow \pi_1(X, z)$ is trivial for some $z \in U$, as X is locally path-connected and semi-locally simply connected, this is a basis for the topology on X . Now define a basis on $\tilde{X}$, for $U \in \mathcal{U}$ and $\gamma : I \rightarrow X$ from x_0 to some $x \in U$, consider $V_{U, [\gamma]} = \{[\gamma * \eta] \mid \eta : I \rightarrow U, \eta(0) = \gamma(1)\}$. We can immediately check: if $[\gamma] = [\gamma']$ then $V_{U, [\gamma]} = V_{U, [\gamma']}$, if $[\gamma'] \in V_{U, [\gamma]}$ then $V_{U, [\gamma]} = V_{U, [\gamma']}$, and if $\gamma(1) \in W \in \mathcal{U}$, then $V_{W, [\gamma]} \subset V_{U, [\gamma]}$, so let these form a basis for the topology on $\tilde{X}$. $\square$

TODO: finish the proof

1.5 Fundamental theorem

Theorem 1.5.1. For (X, x) path-connected, locally path-connected, and semi-locally simply connected, there is a bijection between isomorphism class of covers of (X, x) and subgroup of $\pi_1(X, x)$, by sending p to $p_*\pi_1(Y, y)$.

Proof. We have already shown this is injective, so for a subgroup $G \leq \pi_1(X, x)$ we have to construct a cover $p : (Y, y) \rightarrow (X, x)$ with $p_*\pi_1(Y, y) = G$, let $q : \tilde{X} \rightarrow X$ be the universal cover, and $H \leq \text{Deck}(\tilde{X})$ correspond to G , then let $Y = \tilde{X}/H$ with basepoint $y = [\tilde{x}]$, then for all $h \in H$, we have $q(z) = q(h(z))$, the universal property gives us $p : (Y, y) \rightarrow (X, x)$ with $p\pi = q$ sending $[z]$ to $q(z)$.

To show p is a covering, choose $w \in X$ with $w \in U \subset X$ a path-connected neighbourhood with $q^{-1}(U) = \bigsqcup_{\alpha \in A} V_\alpha \subset \tilde{X}$ with each $q|_{V_\alpha} V_\alpha \rightarrow U$ a homeomorphism. $\square$

TODO: finish this too

Proposition 1.5.2. Any subgroup of a free group is free.

2 Homology

If we want an invariant that detects higher dimensional holes, the natural choice would be to consider $\pi_n(X)$ the homotopy classes of maps $S^n \rightarrow X$; unfortunately, these are very difficult to compute. By adding some combinatorial structure to our shapes, and keeping our groups abelian, we can define a functor that's much easier to compute on many of the nice spaces we care about.

2.1 Simplicial homology

Definition 2.1.1. A Δ -complex structure on a space X is a collection of maps:

$$\sigma_\alpha : \Delta^n \rightarrow X$$

where n depends on α , with all of the σ_α satisfying the following:

- The restriction of each σ_α to $\mathring{\Delta}^n$ is injective;
- Each point $x \in X$ lies in exactly one of the $\sigma_\alpha(\mathring{\Delta}^n)$;
- For each face map δ_i , there exists some σ_β in the Δ -complex with $\sigma_\alpha \circ \delta_i = \sigma_\beta$;
- A subset $U \subset X$ is open iff all of the $\sigma_\alpha^{-1}(U)$ is open in each Δ^n .

Δ -complexes are a special example of CW-complexes, thus are Hausdorff. When considering these examples, it is paramount to conflate the most injective maps σ_α with their images in X , which will be simplices with some collapsing of their boundaries.

TODO:

Definition 2.1.2. Given a Δ -complex structure on X , we form the **simplicial complex**:

$$\dots \xrightarrow{d_{n+1}^\Delta} C_n^\Delta(X) \xrightarrow{d_n^\Delta} C_{n-1}^\Delta(X) \xrightarrow{d_{n-1}^\Delta} \dots \xrightarrow{d_3^\Delta} C_2^\Delta(X) \xrightarrow{d_2^\Delta} C_1^\Delta(X) \xrightarrow{d_1^\Delta} C_0^\Delta(X) \xrightarrow{d_0^\Delta} 0$$

where $C_n^\Delta(X)$ is the free abelian groups on the n -simplices in X , and the differential is given by

$$d_n^\Delta(\sigma_\alpha) = \sum_{i=0}^n (-1)^i \sigma_\alpha \circ \delta_i$$

Proposition 2.1.3. $d_\bullet^\Delta$ does in fact make $C_\bullet^\Delta$ a chain complex, i.e. $d_n^\Delta \circ d_{n+1}^\Delta = 0$ for all n .

Proof. By linearity, it suffices to consider an $(n+1)$ -simplex $\sigma_\alpha : \Delta^{n+1} \rightarrow X$, on which we have:

$$d_n^\Delta d_{n+1}^\Delta(\sigma) = d_n^\Delta \left(\sum_{i=0}^{n+1} (-1)^i \sigma \delta_i \right) = \sum_{i=0}^{n+1} (-1)^i d_n^\Delta(\sigma \delta_i) = \sum_{i=0}^{n+1} (-1)^i \left(\sum_{j=0}^{i-1} (-1)^j \sigma \delta_i \delta_j - \sum_{j=i+1}^{n+1} (-1)^j \sigma \delta_i \delta_j \right)$$

in this sum, for each $0 \leq a < b \leq n+1$, there will be two terms of $\sigma \delta_a \delta_b$, one with coefficient $(-1)^{a+b}$, and the other with $-(-1)^{a+b}$, thus the sum is zero on all $(n-1)$ -simplices. $\square$

Definition 2.1.4. If X has a Δ -complex structure, then we define the **simplicial homology** of X , as the homology of the chain complex $(C_\bullet^\Delta, d_\bullet^\Delta)$, we write this as $H_*^\Delta(X) := H_*(C_\bullet^\Delta(X), d)$.

TODO: computations and pictures

Maps of spaces don't necessarily induce maps on chain complexes, there is no guarantee that the image of the Δ -complex is a Δ -complex.

TODO: give an example

2.2 Singular homology

We want a formulation of homology that is immediately functorial.

Definition 2.2.1. To a space X we can associate the **singular complex** $C_\bullet(X)$, when $C_n(X)$ is the free abelian group on maps $\sigma : \Delta^n \rightarrow X$, and the differential is as with the simplicial complex.

From this we also define the **singular homology** of X , $H_*(X) := H_*(C_\bullet(X), d)$

Proposition 2.2.2. A continuous map $f : X \rightarrow Y$ induces homomorphisms $f_* : H_n(X) \rightarrow H_n(Y)$ for all n , this makes each homology group a functor $H_n(-) : \mathbf{Top} \rightarrow \mathbf{Ab}$.

Proof. We will show f induces a chain map $f_\# : C_\bullet(X) \rightarrow C_\bullet(Y)$. We define $f_\#$ on generators by $f_\#(\sigma) = f \circ \sigma$, to see this is a chain map observe:

$$f_\#(d\sigma) = f_\# \left(\sum_{i=0}^n (-1)^i \sigma \circ d_i \right) = \sum_{i=0}^n (-1)^i f_\#(\sigma \circ d_i) = \sum_{i=0}^n (-1)^i (f \circ \sigma) \circ d_i = d(f \circ \sigma) = d(f_\#)$$

All the other properties are obvious. $\square$

Corollary 2.2.3. If $f : X \rightarrow Y$ is a homeomorphism, then f_* is an isomorphism.

We can go further, showing homotopy equivalences induce isomorphisms on homology.

Proposition 2.2.4. If $f, g : X \rightarrow Y$ are homotopic maps, then the chain maps $f_\#, g_\#$ are chain homotopic,¹ thus $f_* = g_*$.

Proof. Let $H : X \times I \rightarrow Y$ be the homotopy from f to g , then $H \circ (\sigma \times \text{id})$ is a homotopy from $f \circ \sigma$ to $g \circ \sigma$. But the domain of this map, $\Delta^n \times I$, isn't a simplex, we can decompose it as:

$$\Delta^n \times I = [v_0, \dots, v_n] \times I = \bigcup_{i=0}^n [(v_0, 0), \dots, (v_i, 1), (v_i, 1), \dots, (v_n, 1)] =: \bigcup_{i=0}^n G_\sigma^i$$

TODO: low dimensional picture

Giving us $n+1$ n -simplices in Y associated to σ , let's define the obvious maps $K_n : C_n(X) \rightarrow C_{n+1}(Y)$:

$$K_n(\sigma) = \sum_{i=0}^n (-1)^i G_\sigma^i$$

¹Two chain maps $f, g : A_\bullet \rightarrow B_\bullet$ are called **chain homotopic** if there exists some $K : A \rightarrow B[1]$ such that $f - g = dK + Kd$.

which I claim is a chain homotopy. Observe:

$$dK_n(\sigma) = \sum_{i=0}^n (-1)^i d(G_\sigma^i) = \sum_{i=0}^n \left(\sum_{j=0}^i (-1)^{i+j} G_\sigma^i \circ \delta_j - \sum_{j=i}^n (-1)^{i+j} G_\sigma^i \circ \delta_{j+1} \right)$$

$$K_{n-1}(d\sigma) = \sum_{i=0}^n (-1)^i K_{n-1}(\sigma \circ \delta_i) = \sum_{i=0}^n \left(\sum_{j=0}^{i-1} (-1)^{i+j} G_\sigma^j \circ \delta_{i+1} - \sum_{j=i+1}^n (-1)^{i+j} G_\sigma^j \circ \delta_i \right)$$

with some careful inspection, and some picturing the $n = 2$ case, we can see the only terms remaining in the sum are those where we miss a point with the index that G double counts:

$$dK_n(\sigma) + K_{n-1}d(\sigma) = \sum_{i=0}^n (G_\sigma^i \circ \delta_i - G_\sigma^i \circ \delta_{i+1}) = G_\sigma^0 \circ \delta_i - G_\sigma^n \circ \delta_{n+1} = g \circ \sigma - f \circ \sigma$$

Thus, by linearity, $K_\bullet$ is a chain homotopy. $\square$

So we can view each singular homology groups as a functor

$$H_n : \mathbf{hTop} \rightarrow \mathbf{Ab}$$

from the category of topological spaces and *homotopy equivalence classes* of continuous maps.

Corollary 2.2.5. If $f : X \rightarrow Y$ is a homotopy equivalence, then $f_* : H_n(X) \rightarrow H_n(Y)$ is an isomorphism.

2.3 Relative homology

Lemma 2.3.1. Given a short exact sequence of chain complexes:

$$0 \rightarrow A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{j} C_\bullet \rightarrow 0$$

there is a long exact sequence of homology groups:

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \rightarrow \cdots$$

Proof. We first define ∂ :

$$\begin{array}{ccccccc} 0 & \longrightarrow & A_n & \xrightarrow{i} & B_n & \xrightarrow{j} & C_n \longrightarrow 0 \\ & & \downarrow d & & \downarrow d & & \downarrow d \\ 0 & \longrightarrow & A_{n-1} & \xrightarrow{i} & B_{n-1} & \xrightarrow{j} & C_{n-1} \longrightarrow 0 \\ & & \downarrow d & & \downarrow d & & \downarrow d \\ 0 & \longrightarrow & A_{n-2} & \xrightarrow{i} & B_{n-2} & \xrightarrow{j} & C_{n-2} \longrightarrow 0 \end{array} \quad \begin{array}{ccc} & b & \longrightarrow c \\ & \downarrow & \downarrow \\ a & \longrightarrow & d(b) \longrightarrow 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & 0 \end{array}$$

Following the above diagram: given some $[c] \in H_n(C)$, as j is surjective there is some $b \in B_n$ with $j(b) = c$, by commutativity we have $j(d(b)) = d(c) = 0$, so $d(b) \in \ker j = \operatorname{im} i$, as i is injective we can find a unique $a \in A_{n-1}$ with $i(a) = d(b)$, once again by commutativity $i(d(a)) = d(d(b)) = 0$, as i is injective we must have $d(a) = 0$ giving a homology class $[a] \in H_n(A)$ which take as $\partial([c])$.

We have to check our choice is well defined, as we have made a choice for b ; if we chose a different $b' \in B_n$ with $j(b') = c$, then $j(b - b') = 0$, thus $b - b' \in \operatorname{im} i$ with preimage $a' \in A_n$ so $b' = b + i(a')$, now $d(b') = d(b) + di(a') = d(b) + id(a') = i(a + d(a'))$, which will give the same homology class.

I claim ∂ is now obviously a homomorphism $\ker d_n \rightarrow H_{n-1}(A)$.

To show this descends to a map on homology we have to show the image of $\operatorname{im} d \subseteq C_n$ is 0. If $c = d(c')$ for some $c' \in C_{n+1}$, then by surjectivity there exists some $b' \in B_{n+1}$ with $j(b') = c'$, by commutativity our $b = d(b')$ and so $i(a) = d(b) = 0$, thus $a = 0$.

All of the exactness are very direct diagram chases, everything will work if you follow your nose. $\square$

Corollary 2.3.2. For $A \subset X$ a subspace, write $C_\bullet(X, A)$ for $C_\bullet(X)/C_\bullet(A)$, and $H_*(X, A)$ for the homology groups of this complex, then there is a long exact sequence in **relative** homology:

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{\pi_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \rightarrow \cdots$$

Corollary 2.3.3. Given a triple $A \subset B \subset X$, by the third isomorphism theorem there is a long exact sequence in relative homology:

$$\cdots \rightarrow H_n(B, A) \xrightarrow{i_*} H_n(X, A) \xrightarrow{\pi_*} H_n(X, B) \xrightarrow{\partial} H_{n-1}(B, A) \rightarrow \cdots$$

Lemma 2.3.4 (Five lemma). Given the following commutative diagram of abelian groups:

$$\begin{array}{ccccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \xrightarrow{h} & D & \xrightarrow{k} & E \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \downarrow \varepsilon \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \xrightarrow{h'} & D' & \xrightarrow{k'} & E' \end{array}$$

if both rows are exact and $\alpha, \beta, \delta, \varepsilon$ are all isomorphisms, then γ is also an isomorphism.

Proof. We prove this with two different *four lemmas*:

$$(1) \quad \begin{array}{ccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \xrightarrow{h} & D \\ \alpha \downarrow & & \beta \downarrow & & \downarrow \gamma & & \downarrow \delta \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \xrightarrow{h'} & D' \end{array} \quad (2) \quad \begin{array}{ccccccc} B & \xrightarrow{g} & C & \xrightarrow{h} & D & \xrightarrow{k} & E \\ \beta \downarrow & & \gamma \downarrow & & \downarrow \delta & & \downarrow \varepsilon \\ B' & \xrightarrow{g'} & C' & \xrightarrow{h'} & D' & \xrightarrow{k'} & E' \end{array}$$

For (1) we assume both rows are exact, β, δ are injective, and α is surjective; and show γ is injective:

Have $c \in C$ such that $\gamma(c) = 0 \in C'$ so $\delta h(c) = h' \gamma(c) = 0$ and as δ is injective, $h(c) = 0$;
so $c \in \ker h = \text{im } g$, there exists $b \in B$ with $c = g(b)$ and $g'(\beta(b)) = \gamma(g(b)) = \gamma(c) = 0$;
so $\beta(b) \in \ker g' = \text{im } f'$, there exists $a' \in A'$ with $f'(a') = \beta(b)$;
 α surjective $\implies$ there exists $a \in A$ with $a' = \alpha(a)$ and $\beta(b) = f'(\alpha(a)) = \beta(f(a))$;
 β injective $\implies b = f(a)$ so $c = g(b) = f(g(a)) = 0$.

For (2) we assume both rows are exact, β, δ are surjective, and ε is injective; and show γ is surjective:

Have $c' \in C'$; as δ is surjective there is a $d \in D$ with $\delta(d) = h'(c')$;
 $\varepsilon(k(d)) = k' \delta(d) = k' h'(c') = 0$; as ε is injective, $k(d) = 0$;
so $d \in \ker k = \text{im } h$, there exists $c \in C$ with $h(c) = d$; so $h'(\gamma(c)) \delta(h(c)) = \delta(b) = c'$;
so $c' - \gamma(c) \in \ker h' = \text{im } g'$, there is $b' \in B'$ with $g'(b') = c' - \gamma(c)$;
as β is surjective, there is $b \in B$ with $\beta(b) = b'$; so $\gamma(g(b)) = g'(\beta(b)) = c' - \gamma(c)$;
so $c' = \gamma(g(b)) + \gamma(c) = \gamma(g(b) + c)$.

Therefore γ is both injective and surjective, thus an isomorphism. $\square$

All of the previously established properties of homology also hold for relative homology as a functor from the category of pairs of spaces and continuous functions preserving subspaces to **Ab**.

Corollary 2.3.5. Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs. If any two of the follows maps are isomorphisms, the third is as well:

$$(f|_A)_* : H_*(A) \rightarrow H_*(B) \quad f_* : H_*(X) \rightarrow H_*(Y) \quad f_* : H_*(X, A) \rightarrow H_*(Y, B)$$

Proof. Apply the five lemma to the induced maps between the long exact sequences of pairs. $\square$

2.4 Barycentric subdivision

Definition 2.4.1. Let $\mathcal{U} = \{U_i\}$ be a collection of subsets of X whose interiors form an open cover of X . For each n consider

$$C_n^{\mathcal{U}}(X) = \sum_i C_n(U_i) \subseteq C_n(X)$$

equipping this with the usual differential gives us the chain complex $C_\bullet^{\mathcal{U}}(X)$ with the canonical natural map $i : C_\bullet^{\mathcal{U}}(X) \rightarrow C_\bullet(X)$ which induces $i_* : H_n^{\mathcal{U}}(X) \rightarrow H_n(X)$.

With this construction, we want to approximate a general homology class as a sum of small simplices, each of which lie entirely within one component of an open cover. This will be reified with an isomorphism between the $H_*(X)$ and $H_*^{\mathcal{U}}(X)$ above, and our exact method for replacing our simplices with smaller ones will be *barycentric subdivision*.

We define the barycentric subdivision of the n -simplex $\Delta^n = [v_0, \dots, v_n] \subset \mathbb{R}^{n+1}$ inductively. Let $b = \frac{1}{n+1}(v_0 + \dots + v_n)$, then for each facet $[w_1, \dots, w_n]$ of Δ^n , consider the n -simplex $[b, w_1, \dots, w_n]$. This collection of $(n+1)!$ simplices are the barycentric subdivision of Δ^n .

TODO: some pictures in multiple dimensions of multiple iterations of barycentric subdivision.

To show these simplices get appropriately small, recall this lemma from analysis.

Proposition 2.4.2 (Lebesgue covering lemma). Let $\{U_i\}$ be an open cover of a compact metric space X , there exists some $\delta > 0$ such that for all $x \in X$, the ball $B_\delta(x)$ is wholly contained in one of the U_i .

Proof. As X is compact, wlog we may take $\{U_i\}$ to be finite. So we can consider the function $f : X \rightarrow \mathbb{R}$ given by

$$f(x) := \frac{1}{n} \sum_{i=0}^n d(x, U_i^c)$$

as f is continuous and defined on a compact set, it must achieve a minimum δ . And, as every x is contained in some U_i , f is never 0, thus $\delta > 0$. Now, for all $x \in X$, there exists some U_i such that $d(x, U_i^c) \geq \delta$, thus $B_\delta(x) \subseteq U_i$. $\square$

Lemma 2.4.3. Let $\{U_i\}$ be an open cover of the n -simplex Δ^n , then we can barycentrically subdivide Δ finitely many times so that every simplex is wholly contained in one of the U_i .

Proof. Let $\delta > 0$ be the Lebesgue number for this open cover, then every simplex of diameter $< \delta$ will lie whole in one U_i . So it suffices to show the diameter of the subdivided simplices can be made arbitrarily small with a finite number of subdivisions.

Suppose Δ' is a simplex in the barycentric subdivision of Δ , then I claim:

$$\text{diam}(\Delta') \leq \frac{n}{n+1} \text{diam}(\Delta)$$

which is clearly sufficient. The diameter of a linear simplex in $\mathbb{R}^n$ is in fact equal to the distance between a pair of vertices, so for all pairs of vertices $x, y \in \Delta'$ we just have to consider two cases:

- If neither x nor y is the barycenter b , then they belong to a simplex in the barycentric subdivision of a facet of $\tilde{\Delta} \subset \Delta$, so by induction

$$|x - y| \leq \frac{n-1}{n} \text{diam}(\tilde{\Delta}) \leq \frac{n}{n+1} \text{diam}(\tilde{\Delta}) \leq \frac{n}{n+1} \text{diam}(\Delta)$$

- If instead y is the barycenter b , then if we let b_0 be the barycenter of the facet of Δ opposite $x = v_0$, we have

$$|x - y| = \left| v_0 - \frac{v_0 + \dots + v_n}{n+1} \right| = \left| \frac{n}{n+1} v_0 - \frac{n}{n+1} \left(\frac{v_1 + \dots + v_n}{n} \right) \right| = \frac{n}{n+1} |v_0 - b_0| \leq \frac{n}{n+1} \text{diam}(\Delta) \quad \square$$

We now just have to put barycentric subdivision into the form of a chain map. We will do this inductively: for a singular n -simplex $\sigma : \Delta^n \rightarrow X$ consider each of its facets $\sigma \circ \delta_i$, and consider its barycentric subdivision consisting of $(n-1)$ -simplices. Consider the cone of each of these simplices, and send σ to the alternating sums of the subdivision of facets.

$$S(\sigma) = c_b(S(d\sigma))$$